

Premalatha A

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SUMMARY

Software Developer with 1+ year of experience at Tata Consultancy Services working on enterprise application modernization and COBOL-to-Java migration. Experienced in Java development, debugging, application optimization, REST API development, SQL, Git, and backend application development. Hands-on experience with Spring Boot, Spring Data JPA, Hibernate, MySQL, Docker, Maven, Swagger/OpenAPI, and Jenkins fundamentals. Strong foundation in Object-Oriented Programming, Data Structures and Algorithms, DBMS, problem solving, and layered application architecture. Currently developing an Angular frontend for a Java Spring Boot Employee Management application.

EDUCATION

SCAD College of Engineering and Technology
M.E. Computer Science and Engineering

Tirunelveli, Tamil Nadu
2024 – 2026

SCAD College of Engineering and Technology
B.E. Computer Science and Engineering

Tirunelveli, Tamil Nadu
2020 – 2024

EXPERIENCE

Assistant System Engineer
Tata Consultancy Services (TCS)

February 2025 – Present
Chennai, Tamil Nadu

- Contribute to a COBOL-to-Java application modernization project involving the analysis and migration of legacy application modules.
- Develop and maintain Java application components as part of the modernization process.
- Analyze existing COBOL modules and legacy application logic and convert functionality into maintainable Java solutions.
- Perform coding, debugging, troubleshooting, and application optimization to improve software reliability and performance.
- Analyze structured legacy system data and application behavior during the migration process.
- Apply Java, Git, and code review practices while contributing to software development activities.
- Collaborate with team members to analyze application issues and support project deliverables.
- Work with legacy application components while contributing to their transition toward modern Java-based application development.

PROJECTS

Employee Management System | Full-Stack Application | Angular | TypeScript | Java | Spring Boot | REST APIs | Spring Data JPA | Hibernate | MySQL | Docker | Swagger/OpenAPI

- Developing a full-stack Employee Management System with an Angular frontend and Java Spring Boot backend.
- Developed RESTful APIs using Spring Boot to manage employee information through CRUD operations.
- Developing Angular components and services to consume backend REST APIs and display employee data.
- Implementing employee forms and validation for creating and updating employee records.
- Integrating the Angular frontend with Spring Boot REST APIs for frontend-backend communication.
- Using Spring Data JPA and Hibernate for database interaction and object-relational mapping.
- Integrating MySQL for persistent employee data storage.
- Applying layered backend architecture using Controller, Service, Repository, and Entity components.
- Implementing exception handling and validation to improve application reliability and data accuracy.

- Using Swagger/OpenAPI for API documentation, testing, and API exploration.
- Containerizing the backend application using Docker for consistent deployment environments.

Expense & Personal Finance Tracker API | Java | Spring Boot | Spring Data JPA | Hibernate | MySQL | REST APIs

- Developing a Personal Finance Tracker application using Java and Spring Boot to manage income, expenses, budgets, and spending categories.
- Designing and implementing RESTful APIs for financial transaction management and CRUD operations.
- Using Spring Data JPA and Hibernate for database interaction and object-relational mapping.
- Integrating MySQL to store and manage transaction, budget, income, and expense data.
- Implementing monthly expense summaries and budget tracking functionality.
- Applying layered architecture using Controller, Service, Repository, and Entity components.
- Implementing exception handling and validation to improve application reliability and data accuracy.
- Following REST API and backend development best practices.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, TypeScript

Backend Development: Spring Boot, Spring Data JPA, Hibernate, REST APIs

Frontend Development: Angular, HTML, CSS

Database: MySQL

Software Engineering: Object-Oriented Programming, Data Structures and Algorithms, DBMS, Debugging, Exception Handling, Validation, API Testing

Architecture: Layered Architecture, Controller-Service-Repository Architecture, System Design Fundamentals, Microservices Architecture (Conceptual Knowledge)

Tools & DevOps: Git, GitHub, Bitbucket, Docker, Maven, Swagger/OpenAPI, Jenkins (Basics)

Cloud: AWS (Basics)

Development Practices: Code Review, Agile, Problem Solving

CERTIFICATIONS

Claude Architect Foundations — Anthropic

Azure Security Engineer Associate — Microsoft

AI Agents with Claude API in Python — Udemy

Python for Absolute Beginners — Udemy

Generative AI for Beginners — Udemy

ADDITIONAL INFORMATION

Languages: English, Tamil, Malayalam Award: 2nd Place — TCS Office Badminton Tournament, November 2025